

HAT-P-12b

R-1323

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-12_210408 · created 2026-07-31T19:46:17+00:00

BENCHMARK: INACCURATE

Results

Mid-Transit Time (Tmid)	2459312.6894 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.182 +/- 0.01
Transit depth (Rp/Rs)^2	3.33 +/- 0.37 [%]
Orbital Inclination (inc)	88.07 +/- 0.78
Airmass coefficient 1 (a1)	1.186 +/- 0.025
Airmass coefficient 2 (a2)	-0.1462 +/- 0.0094
Scatter in the residuals of the lightcurve fit is	2.14 %
Best Comparison Star	None
Optimal Aperture	4.32
Optimal Annulus	22.94
Transit Duration (day)	0.0956 +/- 0.0055

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.182 $\pm$ 0.01 vs 0.1406 (deviation 4.14 σ)
Duration fit vs published	0.0956 d vs 0.0975 d (deviation 0.35 σ)
Mid-time O–C	8.7 min
Depth SNR	18.2
Residual scatter	2.14 %

Light curve

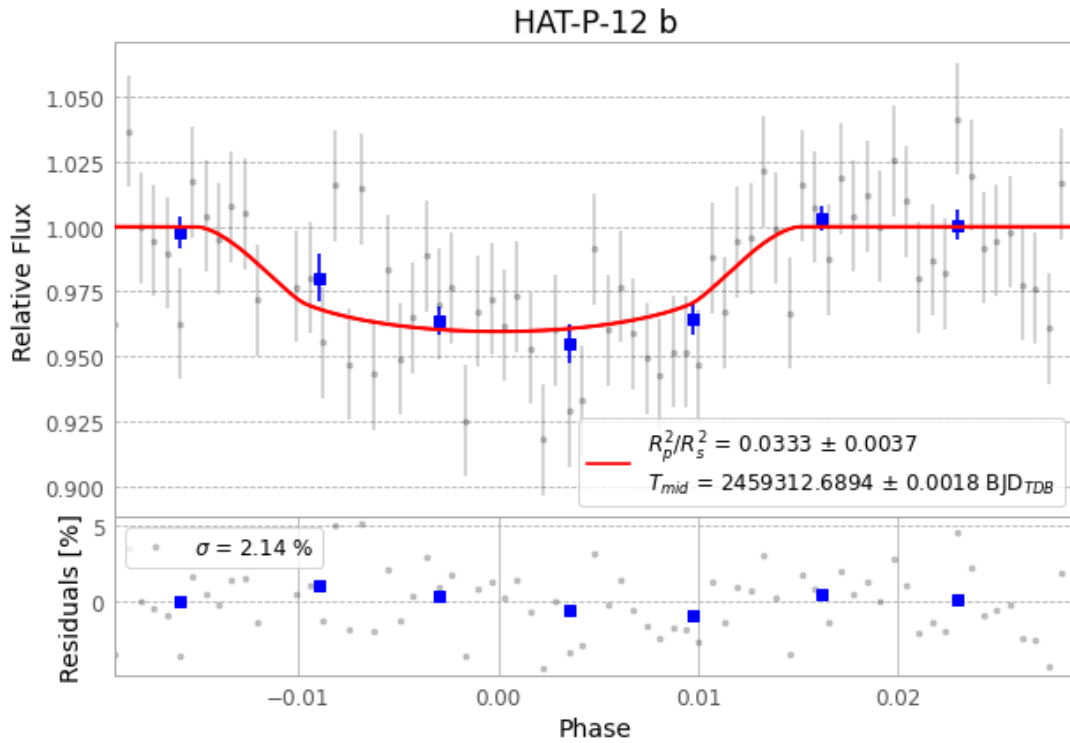


Figure 1 — FinalLightCurve_HAT-P-12 b_2021-04-07.png (SHA-256 b340daa60fb4166b...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [290, 269], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 1a34f5cf4c99cfa6...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

73 FITS frames (48 MB), HATP-12210408025712.FITS → HATP-12210408063912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-12-210408.log (304e0901fce2...), FinalLightCurve_HAT-P-12 b_2021-04-07.png (b340daa60fb4...), FinalLightCurve_HAT-P-12 b_2021-04-07.csv (2c43d94209c4...)

Compute resources

Wall time 160.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)